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(54) **X-RAY SENSITIVE BATTERY SEPARATORS AND RELATED METHODS**

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(57) **ABSTRACT**

The instant application relates to an X-ray sensitive battery separator for a secondary lithium battery and a method for detecting the position of a separator in a secondary lithium battery. The X-ray sensitive battery separator includes a microporous membrane having an X-ray detectable element therein, thereon, or added thereto. The X-ray detectable element constitutes less than 20% by weight of the microporous membrane or separator. The method for detecting the position of a separator in a battery, cell, stack, jellyroll, can, or the like includes the following steps: (1) providing a battery, cell, stack, jellyroll, or the like including an X-ray sensitive battery separator; (2) subjecting the battery, cell, stack, jellyroll, or the like to X-ray radiation; and (3) thereby detecting the position of said separator in said battery, cell, stack, jellyroll, or the like.

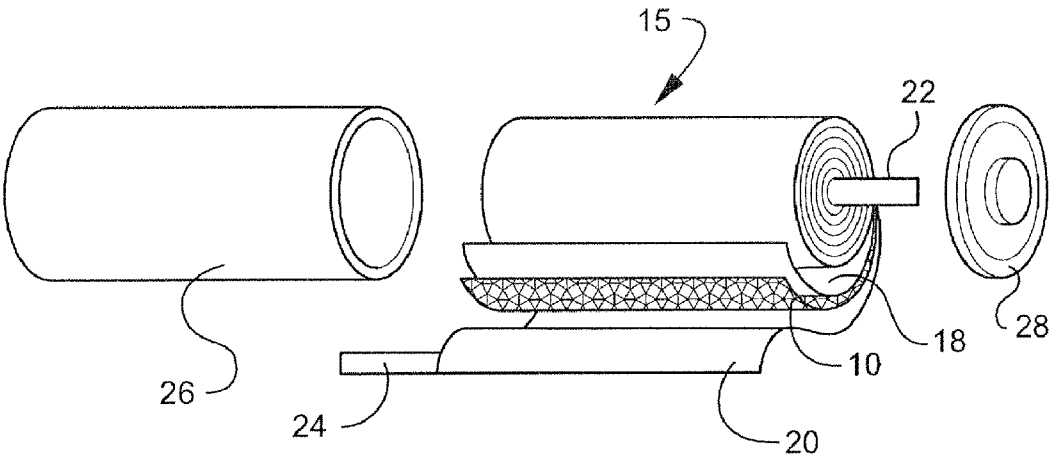
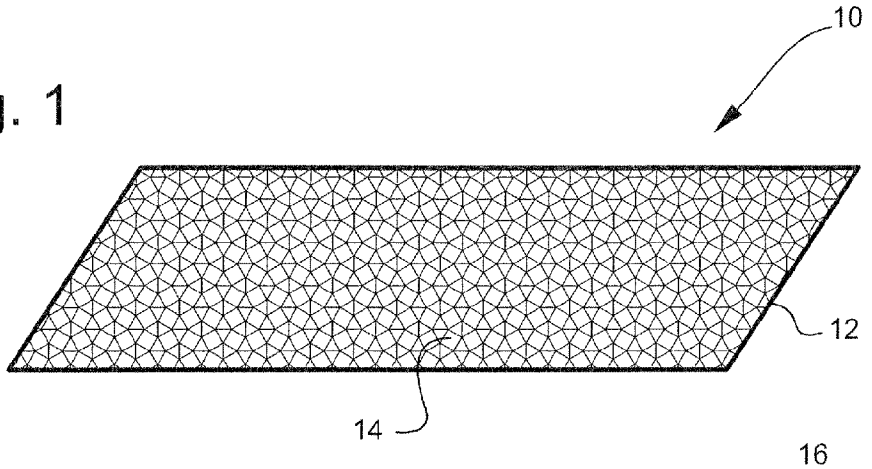


Fig. 1



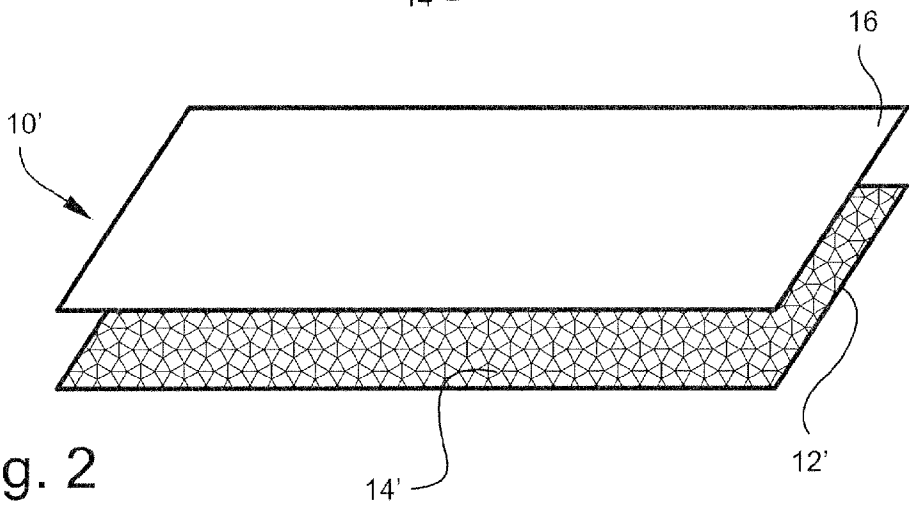
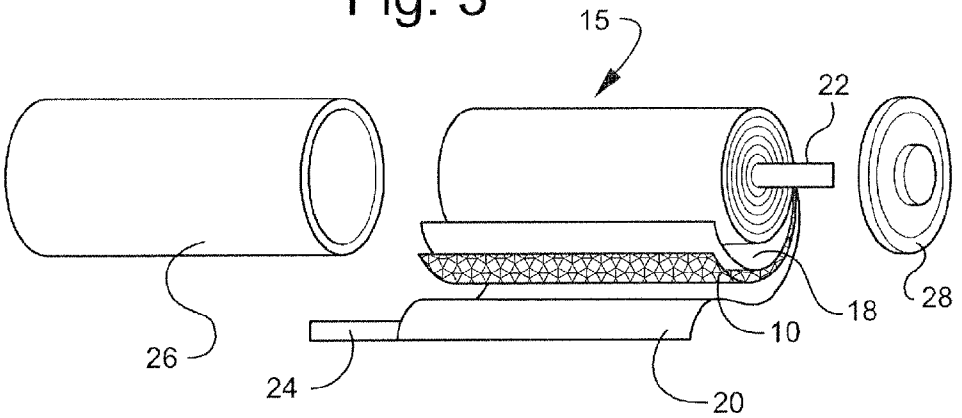


Fig. 2

Fig. 3



X-RAY SENSITIVE BATTERY SEPARATORS AND RELATED METHODS

FIELD OF INVENTION

[0001] The instant application relates to X-ray sensitive or detectable battery separators and methods for making and using such separators, including methods for detecting the position of such a separator in a battery, cell, stack, jellyroll, can, or the like.

BACKGROUND OF THE INVENTION

[0002] A battery separator is used to separate the positive and negative electrodes of a battery, for example, in a secondary lithium battery. A battery separator is typically microporous to allow ionic current with least possible resistance while preventing the electrodes from direct contact resulting in an internal short.

[0003] In general, a battery separator is sandwiched between the positive electrode and the negative electrode of a secondary lithium battery. It is important for a battery separator to remain in its proper position because even a minute displacement may cause a short in the battery. Currently, other than as described in US Publication US2009/0081535 A1, published Mar. 26, 2009, there are no prevailing techniques to determine the position of a separator in a battery to prevent the introduction of flawed batteries, i.e. those batteries in which the battery separator (or electrode) was displaced during the manufacturing process, into the consumer market.

[0004] Microporous polymer membranes are known, can be made by various processes, and the process by which the membrane is made may have an impact upon the membrane's physical attributes. See, for example, Kesting, Robert E., *Synthetic Polymeric Membranes, A Structural Perspective,* Second Edition, John Wiley & Sons, New York, NY, (1985). Three different known processes for making microporous polymer membranes include: the dry-stretch process (also known as the CELGARD process), the wet process, and the particle stretch process.

[0005] The dry-stretch process (the CELGARD process) refers to a process where pore formation results from stretching a nonporous, semicrystalline, extruded polymer precursor in the machine direction (MD stretch). See, for example, Kesting, Ibid. pages 290-297, incorporated herein by reference. Such a dry-stretch process is different from the wet process and the particle stretch process. Generally, in the wet process, also known as the phase inversion process, the extraction process, or the TIPS process, the polymeric raw material is mixed with a processing oil (sometimes referred to as a plasticizer), this mixture is extruded, and pores are then formed when the processing oil is removed (these films may be stretched before or after the removal of the oil). See, for example, Kesting, Ibid. pages 237-286, incorporated herein by reference.

[0006] Generally, in the particle stretch process, the polymeric raw material is mixed with a pore formation particulate, this mixture is extruded, and pores are formed during stretching when the interfaces between the polymer and the particulate fracture due to the stretching forces. See, for example, U.S. Pat. Nos. 6,057,061 and 6,080,507, each incorporated herein by reference.

[0007] Moreover, the membranes arising from these different formation processes are usually physically different and the process by which each is made typically distinguishes one membrane from the other. For example, dry-stretch process membranes may have slit shaped pores due to the stretching of the precursor in the machine direction (MD stretch). Wet process membranes tend to have rounder pores and a lacelike appearance due to the oil or plasticizer and the stretching of the precursor in the machine direction (MD stretch) and in the transverse machine direction or transverse direction (TD stretch). Particle stretch process membranes, on the other hand, may have oval shaped pores as the particulate and machine direction stretching (MD stretch) tend to form the pores. Accordingly, each membrane may be distinguished from the other by its method of manufacture.

[0008] While membranes made by the dry-stretch process have met with excellent commercial success, such as a variety of Celgard® dry-stretch porous membranes sold by Celgard, LLC of Charlotte, N.C., including flat sheet membranes, battery separators, hollow fibers, and the like, there is a need to improve, modify or enhance at least selected physical attributes thereof, so that they may be used in a wider spectrum of applications, may perform better for particular purposes, or the like.

[0009] A modified dry-stretch process (modified CELGARD process) involving the formation of unique round shaped pores by, for example, stretching a nonporous, semicrystalline, extruded polymer precursor in the machine direction (MD stretch), followed by stretching in the transverse direction (TD stretch) with machine direction relax (MD relax) is disclosed in US Published Application US2007/0196638 A1, published Aug. 23, 2007, and incorporated by reference herein.

[0010] Despite the research efforts in developing battery separators, there may still be a need for an improved battery separator, such as a battery separator which is x-ray sensitive or readily detectable when inserted or embedded in a battery, cell, stack, jellyroll, can, or the like, to determine its position within the battery, cell, stack, jellyroll, can, or the like, or relative to the electrodes, which is relatively easy to manufacture, is low cost, meets performance requirements, meets product specifications, or the like. Furthermore, there may still be a need for a method for detecting the position of a separator in a battery, cell, stack, jellyroll, can, or the like to determine its position within the battery, cell, stack, jellyroll, can, or the like, or relative to the electrodes, which is relatively easy and cost effective, a method for manufacturing such a separator which is relatively simple and cost effective, a method for using such a separator which is relatively simple and cost effective, or the like.

SUMMARY OF THE INVENTION

[0011] In accordance with at least selected embodiments, the instant application relates to an X-ray sensitive battery separator for a secondary lithium battery and a method for detecting the position of such a separator in a secondary lithium battery. In accordance with at least selected embodiments, the preferred X-ray sensitive battery separator includes a microporous membrane having an X-ray detectable element. In accordance with at least selected embodiments, the X-ray detectable element constitutes a sufficient amount to detect the separator relative to the electrodes (for example, to provide minimum contrast in the x-ray picture). In accordance with at least particular separator embodiments, the X-ray detectable element constitutes less than 20% by weight of the microporous membrane, preferably less than 15% by weight of the microporous membrane, more prefer-

ably less than 10% by weight of the microporous membrane, and most preferably less than 5% by weight of the microporous membrane.

[0012] At least an exemplary method for detecting the position of a separator in a battery, cell, stack, jellyroll, can, or the like, includes the following steps: (1) providing a battery, cell, stack, jellyroll, can, or the like including an X-ray sensitive or detectable battery separator; (2) subjecting the battery, cell, stack, jellyroll, can, or the like to X-ray radiation; and (3) thereby detecting the position of the separator in the battery, cell, stack, jellyroll, can, or the like.

[0013] In accordance with at least selected embodiments of the present invention, there are provided improved battery separators, methods, or the like, such as an improved battery separator which is x-ray sensitive or readily detectable when inserted or embedded in a battery, cell, stack, jellyroll, can, or the like, to determine its position within the battery, cell, stack, jellyroll, can, or the like, or to determine its position relative to the electrodes, which is relatively easy to manufacture, is low cost, meets performance requirements, meets product specifications, and/or the like. Furthermore, in accordance with at least selected embodiments of the present invention, there are provided improved methods for making, using, or detecting the position of a separator in a battery, cell, stack, jellyroll, can, or the like, such as an improved battery separator which is x-ray sensitive or readily detectable, to determine its position within the battery, cell, stack, jellyroll, can, or the like, or relative to the electrodes, which is relatively easy and cost effective, for manufacturing such a separator which is relatively simple and cost effective, for using such a separator which is relatively simple and cost effective, and/or the like.

[0014] In accordance with at least certain embodiments, the instant application relates to an X-ray sensitive battery separator for a secondary lithium battery and a method for detecting the position of such a separator in a secondary lithium battery. The X-ray sensitive battery separator preferably includes a microporous membrane having an X-ray detectable element. In at least one embodiment, the X-ray detectable element, such as barium sulfate particles, preferably constitutes less than 5% by weight of the microporous membrane. The method for detecting the position of such a separator in a battery includes the following steps: (1) providing a battery including an X-ray sensitive battery separator; (2) subjecting the battery to X-ray radiation; and (3) thereby detecting the position of said separator in said battery.

[0015] At least selected embodiments of the present invention relate to dry-stretch X-ray sensitive or detectable battery separators and to dry-stretch methods for making and methods of using such separators, including methods for detecting the position of such a separator in a battery, cell, stack, jellyroll, can, or the like.

[0016] At least selected embodiments of the present invention relate to wet process X-ray sensitive or detectable battery separators and to wet process methods for making and methods of using such separators, including methods for detecting the position of such a separator in a battery, cell, stack, jellyroll, can, or the like.

[0017] At least selected embodiments of the present invention relate to particle stretch X-ray sensitive or detectable battery separators and to particle stretch methods for making and methods of using such separators, including methods for detecting the position of such a separator in a battery, cell, stack, jellyroll, can, or the like.

[0018] At least selected embodiments of the present invention relate to modified dry-stretch X-ray sensitive or detectable battery separators and to modified dry-stretch methods for making and methods of using such separators, including methods for detecting the position of such a separator in a battery, cell, stack, jellyroll, can, or the like.

BRIEF DESCRIPTION OF THE DRAWINGS

[0019] For the purpose of illustrating the invention, there is shown in the drawings a form that is presently preferred; it being understood, however, that this invention is not limited to the precise embodiments, arrangements and instrumentalities shown.

[0020] FIG. **1** is a schematic perspective view of a first embodiment of an X-ray sensitive battery separator according to the instant invention;

[0021] FIG. **2** is a schematic perspective view of second embodiment of an X-ray sensitive battery separator according to the instant invention; and

[0022] FIG. **3** is an exploded view of a battery or can including the X-ray sensitive battery separator of FIG. **1**.

DETAILED DESCRIPTION OF THE INVENTION

[0023] Referring to the drawings wherein like numerals indicate like elements, there is shown, in FIG. **1**, a first embodiment of an X-ray sensitive battery separator **10**. The X-ray sensitive battery separator **10** includes a microporous membrane **12**, which contains an X-ray detectable element **14** dispersed therethrough.

[0024] Microporous membrane **12** may be any microporous membrane which contains X-ray detectable element **14**. Microporous membranes are generally known in the art. Microporous membrane **12** may be made from any material, for example a polymer. A polymer, for example, may be any synthetic polymer, cellulose, or synthetically modified cellulose. The preferred synthetic polymers are polyolefins, e.g., polyethylene (PE), polypropylene (PP), polymethylpentene, polybutylene, ultra high molecular weight polyethylene, ultra high molecular weight polypropylene, copolymers thereof, and mixtures or blends thereof. Microporous membrane **12** may have any porosity; for example, microporous membrane **12** may have a porosity in the range of about 20% to about 80%. Microporous membrane **12** may have any average pore size; for example, microporous membrane **12** may have an average pore size in the range of about 0.1 micron to about 5 microns. Microporous membrane **12** may be made of one or more plies and may have any thickness; for example, microporous membrane **12** may have a thickness in the range of about 6 microns to about 80 microns.

[0025] X-ray detectable element **14** may be any X-ray detectable material. For example, X-ray material **14** may be a material selected from the group consisting of a metal oxide, a metal phosphate, a metal carbonate, an X-ray fluorescent material, a metal sulfate or salt such as barium sulfate ($BaSO_4$), and combinations thereof. The listed X-ray materials are not limiting. Exemplary metal oxides include, but are not limited to, metal oxides having a metal selected from the group consisting of Zn, Ti, Mn, Ba, Ni, W, Hg, Si, Cs, Sr, Ca, Rb, Ta, Zr, Al, Pb, Sn, Sb, Cu, Ni, and Fe. The listed metal oxides are not limiting. Exemplary metal phosphates include, but are not limited to, phosphate oxides having a metal selected from the group consisting of Zn, Ti, Mn, Ba, Ni, W, Hg, Si, Cs, Sr, Ca, Rb, Ta, Zr, Al, Pb, Sn, Sb, Cu, Ni, and Fe.

The listed metal phosphates are not limiting. Exemplary metal carbonates include, but are not limited to, metal carbonates having a metal selected from the group consisting of Zn, Ti, Mn, Ba, Ni, W, Hg, Si, Cs, Sr, Ca, Rb, Ta, Zr, Al, Pb, Sn, Sb, Cu, Ni, and Fe. The listed metal carbonates are not limiting. Exemplary X-ray fluorescent materials include, but are not limited to, organic materials, inorganic materials, and combinations thereof. A fluorescent material, as used herein, refers to a material having electrons capable of becoming exited by X-ray radiation thereby providing detection signals. The listed X-ray fluorescent materials are not limiting. X-ray detectable element **14** may constitute any percentage of the weight of membrane **12**. For example, the X-ray detectable element may constitute in the range of 0.01 to 98 percent by weight of the membrane **12**, possibly preferably less than 20 percent by weight of the membrane **12**, more preferably less than 15 percent by weight of the membrane **12**, and most preferably less than 10 percent by weight of the membrane **12**. When barium sulfate particles are used as an x-ray detectable element, in one possibly preferred embodiment, the barium sulfate is less than 10 percent by weight of the membrane **12**, possibly more preferably between 2 and 5 percent by weight of the membrane **12**, and possibly most preferably about 4 percent by weight of the membrane **12**.

[0026] In the alternative, referring to FIG. **2**, the X-ray sensitive battery separator **10'** may be a multi-layer battery separator. Multi-layer, as used herein, refers to two or more layers. The X-ray detectable battery separator **10'** preferably includes at least one microporous membrane or layer **12'**, which contains an X-ray detectable element **14'**, and at least one other porous membrane, material or layer **16**. Preferably, the X-ray sensitive or detectable battery separator **10'** includes a plurality of layers **16**, for example, one on each side of layer **12'**. In one embodiment, the layer **16** is an additional layer **12'**. Further, at least one of layers **16** or **12'** or an additional layer may be a shutdown layer, i.e., one adapted to shut down ionic flow between the electrodes in the event of thermal runaway or internal short circuiting caused by internal or external circumstances.

[0027] Microporous membrane or layer **12'** (or layers **12'**) may be any microporous membrane which contains X-ray detectable element **14'**. Microporous membranes are generally known in the art. Microporous membrane **12'** may be made from any material, for example a polymer. A polymer, for example, may be any synthetic polymer, cellulose, or synthetically modified cellulose. The preferred synthetic polymers are polyolefins, e.g., polyethylene (PE), polypropylene (PP), polymethylpentene, polybutylene, ultra high molecular weight polyethylene (UHMWPE), ultra high molecular weight polypropylene (UHMWPP), copolymers thereof, and mixtures or blends thereof. Microporous membrane **12'** may have any porosity; for example, microporous membrane **12'** may have a porosity in the range of about 20% to about 80%. Microporous membrane **12'** may have any average pore size; for example, microporous membrane **12'** may have an average pore size in the range of about 0.1 micron to about 5 microns. Microporous membrane **12'** may be made of one or more plies and have any thickness; for example, microporous membrane **12'** may have a thickness in the range of about 6 microns to about 80 microns.

[0028] X-ray detectable element **14'** (like element **14**) may be any X-ray detectable material. For example X-ray material **14'** may be a material selected from the group consisting of a metal oxide, a metal phosphate, a metal carbonate, and an X-ray fluorescent material, a metal sulfate or salt such as barium sulfate ($BaSO_4$), and combinations thereof. The listed X-ray sensitive or detectable materials are not limiting. Exemplary metal oxides include, but are not limited to, metal oxides having a metal selected from the group consisting of Zn, Ti, Mn, Ba, Ni, W, Hg, Si, Cs, Sr, Ca, Rb, Ta, Zr, Al, Pb, Sn, Sb, Cu, Ni, and Fe. The listed metal oxides are not limiting. Exemplary metal phosphates include, but are not limited to, phosphate oxides having a metal selected from the group consisting of Zn, Ti, Mn, Ba, Ni, W, Hg, Si, Cs, Sr, Ca, Rb, Ta, Zr, Al, Pb, Sn, Sb, Cu, Ni, and Fe. The listed metal phosphates are not limiting. Exemplary metal carbonates include, but are not limited to, metal carbonates having a metal selected from the group consisting of Zn, Ti, Mn, Ba, Ni, W, Hg, Si, Cs, Sr, Ca, Rb, Ta, Zr, Al, Pb, Sn, Sb, Cu, Ni, and Fe. The listed metal carbonates are not limiting. Exemplary X-ray fluorescent materials include, but are not limited to, organic materials, inorganic materials, and combinations thereof. A fluorescent material as used herein refers to a material having electrons capable of becoming exited by X-ray radiation thereby providing detection signals. The listed X-ray fluorescent materials are not limiting. X-ray detectable element **14'** may constitute any percentage of the weight of membrane **12'**. For example, the X-ray detectable element **14'** may constitute in the range of 0.01 to 98 percent by weight of the separator **10'** or membrane **12'**, possibly preferably less than 20 percent by weight of the membrane **12'**, more preferably less than 15 percent by weight of the membrane **12'**, and most preferably less than 10 percent by weight of the membrane **12'**. When barium sulfate particles are used as an x-ray detectable element, in one possibly preferred embodiment, the barium sulfate is less than 10 percent by weight of the membrane **12'**, possibly more preferably between 2 and 5 percent by weight of the membrane **12'**, and possibly most preferably about 4 percent by weight of the membrane **12'**.

[0029] Layer or layers **16** may be any conventional porous or microporous membrane, material or layer. Porous or microporous membranes or materials are generally known in the art. Layer or layers **16** may be made from any material, for example a polymer. A polymer, for example, may be any synthetic polymer, cellulose, or synthetically modified cellulose. The preferred synthetic polymers are polyolefins, e.g., polyethylene (PE), polypropylene (PP), polymethylpentene, polybutylene, ultra high molecular weight polyethylene, ultra high molecular weight polypropylene, copolymers thereof, and mixtures or blends thereof. Layer **16** may have any porosity; for example, layer **16** may have a porosity in the range of about 20% to about 80%. Layer **16** may have any average pore size; for example, layer **16** may have an average pore size in the range of about 0.1 micron to about 5 microns. Layer **16** may be made of one or more plies and have any thickness; for example, layer **16** may have a thickness in the range of about 10 microns to about 40 microns. When separator **10'** includes more than one layer **16**, each such layer **16** may be of the same or different construction. As non-limiting examples, separator **10'** may have a PP layer **12'** and a PP layer **16**, a PP layer **12'** and a PE layer **16**, a PE layer **12'** and a PP layer **16**, a PE layer **12'** and a PE layer **16**, a PE layer **12'** between two like PP layers **16**, a PE layer **12'** between two different PP layers **16**, a PP layer **12'** between two like PE layers **16**, a PP layer **12'** between two different PE layers **16**, a PE layer **12'** between a first PP layer **16** and a second PE layer **16**, two PP layers **12'**, two PP layers **12'** and a PE layer **16**, two PE layers **12'**, two PE layers **12'** between two PP layers **16**, or the like. Layers **12'**

and **16** of separator **10'**, may, for example, be coextruded, laminated, or bonded together.

[0030] In manufacturing, referring to FIG. **3**, the X-ray sensitive battery separator **10** of FIG. **1** (or **10'** of FIG. **2**), is sandwiched between a positive electrode **18** and a negative electrode **20**, and may be subsequently rolled into a jellyroll **15** (prismatic constructions or stacks, rectangular cells, pocket cells, button cells, other cans, containers, constructions, or the like, are also possible). The jellyroll **15** may further include negative tab **24**, and positive tab **22**. Positive electrode **18** may include a metal sheet, e.g., aluminum foil, i.e. the current collector, upon which the positive electrode material or electrode active mix (not shown but conventional) has been spread in conventional manner. Negative electrode **20** may include a metal sheet, e.g., copper foil, i.e. the current collector, upon which the negative electrode material or electrode active mix (not shown but conventional) has been spread in conventional manner. Subsequently, jellyroll **15** is inserted into can **26**, which is filled with an electrolyte (not shown), and then, can **26** is sealed with cap **28** (or with a cap at each end). Can **26** may be a metallic (e.g., steel, stainless steel, aluminum) cylindrical can, a plastic box, of a foil (e.g., metallized foil) pouch, or the like. Electrolyte may be any substance capable of providing ionic conductivity. Electrolyte may, for example, be a liquid electrolyte, a solid electrolyte, or a polymer or gel electrolyte. A liquid electrolyte generally includes an electrolytic salt dissolved in a solvent, i.e. an inorganic solvent or an organic solvent. A gel electrolyte generally includes an electrolytic salt dissolved in non-aqueous solvent, and gelated with a polymer matrix.

[0031] In operation, a battery, cell, stack, jellyroll, can, or the like containing an X-ray sensitive battery separator **10** (or **10'**) is subjected to X-ray radiation thereby facilitating the detection of the position of the X-ray sensitive battery separator **10** (or **10'**) within the battery, cell, stack, jellyroll, can, or the like. For example, the separator is usually wider than the electrodes, so that the separator extends beyond the lateral edges of the electrodes. The separator separates and extends beyond the lateral edges of the electrodes to prevent the electrodes from coming into physical contact and thereby creating the potential for short-circuiting. It is possible that during winding or in the battery assembly that the separator portion that extends beyond the lateral edges of the electrodes is removed, shifted or pushed back or otherwise misplaced (or the electrodes are moved or misplaced) thereby allowing the possibility of physical contact of the electrodes. An X-ray examination of the assembled battery, cell, stack, or the like allows a check, inspection or test to determine that the separator remains (or electrodes remain) in position throughout manufacture. The X-ray visible separator can be observed, via X-ray examination, to ensure that it has maintained its position (i.e., a portion extending beyond the lateral edges of the electrodes). Further, it is possible that the x-ray inspection process could be automated, via computer, to increase the speed of inspection.

[0032] In accordance with at least selected possibly preferred embodiments, the membranes or layers **12**, **12'**, and/or **16** are made by the dry-stretch process (the CELGARD process) where pore formation results from stretching a nonporous, semicrystalline, extruded polymer precursor in the machine direction (MD stretch). See, for example, Kesting, Ibid. pages 290-297, incorporated herein by reference. Such a dry-stretch process is different from the wet process and the particle stretch process.

[0033] In accordance with at least other selected possibly preferred embodiments, the membranes or layers **12**, **12'**, and/or **16** are made by the wet process, also known as the phase inversion process, the extraction process, or the TIPS process, where the polymeric raw material is mixed with a processing oil (sometimes referred to as a plasticizer), this mixture is extruded, and pores are then formed when the processing oil is removed (these films may be stretched before or after the removal of the oil). See, for example, Kesting, Ibid. pages 237-286, incorporated herein by reference.

[0034] In accordance with at least still other selected possibly preferred embodiments, the membranes or layers **12**, **12'**, and/or **16** are made by the particle stretch process, where the polymeric raw material is mixed with pore forming particulate, this mixture is extruded, and pores are formed during stretching when the interfaces between the polymer and the particulate fracture due to the stretching forces. See, for example, U.S. Pat. Nos. 6,057,061 and 6,080,507, each incorporated herein by reference.

[0035] In accordance with at least yet other selected possibly preferred embodiments, the membranes or layers **12**, **12'**, and/or **16** are made by a modified dry-stretch process (modified CELGARD process) involving, for example, stretching a nonporous, semicrystalline, extruded polymer precursor in the machine direction (MD stretch), followed by stretching in the transverse direction (TD stretch) with machine direction relax (MD relax). See, for example, US Published Application US2007/0196638 A1, published Aug. 23, 2007, and incorporated by reference herein.

[0036] In accordance with at least still yet other selected embodiments, the membranes or layers **12**, **12'**, and/or **16** are polypropylene microporous membranes, made from a beta-nucleated precursor, or beta-nucleated polypropylene (BNPP) as disclosed, for example, in U.S. Pat. No. 6,368,742, incorporated herein by reference. A beta-nucleating agent for polypropylene is a substance that causes the creation of beta crystals in polypropylene.

[0037] Further, in place of or in addition to the x-ray detectable element **14** or **14'** being incorporated into the membrane or layer **12**, **12'**, or **16**, by, for example, being mixed with the polymer prior to extrusion or formation of the precursor, membrane, film, or the like, the x-ray detectable element **14** or **14'** may be applied to the membrane or layer **12**, **12'**, or **16**, to the precursor of membrane or layer **12**, **12'**, or **16**, may be coated on the membrane or layer **12**, **12'**, or **16**, may be applied to the membrane or layer **12**, **12'**, or **16**, or the like. For example, but not limited to, barium sulfate particles may be incorporated into the membrane or layer **12**, **12'**, or **16**, by being mixed with the polymer mix prior to extrusion or formation of the precursor, membrane, film, or the like, may be applied to the membrane or layer **12**, **12'**, or **16**, or to the precursor of membrane or layer **12**, **12'**, or **16**, may be coated on the membrane or layer **12**, **12'**, or **16**, or may be coated on the precursor of the membrane or layer **12**, **12'**, or **16**, or the like. In this way, the x-ray detectable element or elements **14** or **14'** may be incorporated into, on the surface of, applied to, and/or in the pores of the membrane or layer **12**, **12'**, and/or **16**. For example, but not limited to, the X-ray detectable element **14** or **14'** may be incorporated into, on the surface of, applied to, and/or in the pores of the membrane or layer **12** or **12'**, and may constitute any percentage of the weight of membrane **12** or **12'** or of separator **10** or **10'**. For example, the X-ray detectable element **14** or **14'** may constitute in the range

of 0.01 to 98 percent by weight of the separator **10** or **10'** or membrane **12** or **12'**, preferably less than 20 percent by weight of the membrane **12** or **12'**, more preferably less than 15 percent by weight of the membrane **12** or **12'**, and most preferably less than 10 percent by weight of the membrane **12** or **12'**. When barium sulfate particles are used as an x-ray detectable element, in one possibly preferred embodiment, the barium sulfate is less than 10 percent by weight of the separator **10** or **10'** or of the membrane **12** or **12'**, possibly more preferably between 2 and 5 percent by weight of the membrane **12** or **12'**, and possibly most preferably about 4 percent by weight of the membrane **12** or **12'**. When barium sulfate particles are used as an x-ray detectable element, in another possibly preferred embodiment, the barium sulfate is less than 15 percent by weight of the precursor of the membrane **12** or **12'**, possibly more preferably less than 10 percent by weight of the precursor, possibly preferably between 1 and 10 percent by weight of the precursor, and possibly most preferably about 7 to 8 percent by weight of the precursor.

[0038] In one possible example, a porous polymer membrane **12** has a coating including barium sulfate particles and a binder on one surface thereof. The weight percent of barium sulfate is preferably less than 20% by weight of the combined membrane and coating weight.

[0039] In accordance with at least selected embodiments of the present invention, there is provided an X-ray sensitive battery separator for a secondary lithium battery and a method for detecting the position of such a separator in a secondary lithium battery. In accordance with at least selected embodiments, the preferred X-ray sensitive battery separator includes a microporous membrane having an X-ray detectable element. In accordance with at least selected embodiments, the X-ray detectable element constitutes a sufficient amount to detect the separator relative to the electrodes (for example, to provide minimum contrast in the x-ray picture). In accordance with at least particular separator embodiments, the X-ray detectable element constitutes less than 20% by weight of the microporous membrane, preferably less than 15% by weight of the microporous membrane, more preferably less than 10% by weight of the microporous membrane, and most preferably less than 5% by weight of the microporous membrane.

[0040] At least an exemplary method for detecting the position of a separator in a battery, cell, stack, jellyroll, can, or the like, includes the following steps: (1) providing a battery, cell, stack, jellyroll, can, or the like including an X-ray sensitive or detectable battery separator; (2) subjecting the battery, cell, stack, jellyroll, can, or the like to X-ray radiation; and (3) thereby detecting the position of the separator in the battery, cell, stack, jellyroll, can, or the like.

[0041] In accordance with at least selected embodiments of the present invention, there are provided improved battery separators, methods, or the like, such as an improved battery separator which is x-ray sensitive or readily detectable when embedded in a battery, cell, stack, jellyroll, can, or the like, to determine its position within the battery, cell, stack, jellyroll, can, or the like, or to determine its position relative to the electrodes, which is relatively easy to manufacture, is low cost, meets performance requirements, meets product specifications, and/or the like. Furthermore, in accordance with at least selected embodiments of the present invention, there are provided improved methods for making, using, or detecting the position of a separator in a battery, cell, stack, jellyroll, can, or the like, such as an improved battery separator which is x-ray sensitive or readily detectable, to determine its position within the battery, cell, stack, jellyroll, can, or the like, or relative to the electrodes, which is relatively easy and cost effective, for manufacturing such a separator which is relatively simple and cost effective, for using such a separator which is relatively simple and cost effective, and/or the like.

[0042] In accordance with at least certain embodiments, the instant application relates to an X-ray sensitive battery separator for a secondary lithium battery and a method for detecting the position of such a separator in a secondary lithium battery. The X-ray sensitive battery separator preferably includes a microporous membrane having an X-ray detectable element. The X-ray detectable element, such as barium sulfate particles, preferably constitutes less than 5% by weight of the microporous membrane. The method for detecting the position of such a separator in a battery includes the following steps: (1) providing a battery including an X-ray sensitive battery separator; (2) subjecting the battery to X-ray radiation; and (3) thereby visually detecting the position of said separator in said battery.

[0043] At least selected embodiments of the present invention relate to dry-stretch X-ray sensitive or detectable battery separators and to dry-stretch methods for making and methods of using such separators, including methods for detecting the position of such a separator in a battery, cell, stack, jellyroll, can, or the like.

[0044] At least selected embodiments of the present invention relate to wet process X-ray sensitive or detectable battery separators and to wet process methods for making and methods of using such separators, including methods for detecting the position of such a separator in a battery, cell, stack, jellyroll, can, or the like.

[0045] At least selected embodiments of the present invention relate to particle stretch X-ray sensitive or detectable battery separators and to particle stretch methods for making and methods of using such separators, including methods for detecting the position of such a separator in a battery, cell, stack, jellyroll, can, or the like.

[0046] At least selected embodiments of the present invention relate to modified dry-stretch X-ray sensitive or detectable battery separators and to modified dry-stretch methods for making and methods of using such separators, including methods for detecting the position of such a separator in a battery, cell, stack, jellyroll, can, or the like.

[0047] At least selected embodiments relate to an X-ray sensitive battery separator for a secondary lithium battery and a method for detecting the position of a separator in a secondary lithium battery, an X-ray sensitive battery separator including at least one microporous membrane having an X-ray detectable element therein, thereon, or added thereto, an X-ray detectable element constituting less than 20% by weight of the microporous membrane or separator, and/or a method for detecting the position of a separator in a battery, cell, stack, jellyroll, can, or the like includes the following steps: (1) providing a battery, cell, stack, jellyroll, or the like including an X-ray sensitive battery separator; (2) subjecting the battery, cell, stack, jellyroll, or the like to X-ray radiation; and (3) thereby detecting the position of said separator in said battery, cell, stack, jellyroll, or the like.

[0048] The present invention may be embodied in other forms without departing from the spirit and the essential attributes thereof, and, accordingly, reference should be made to the appended claims, rather than to the foregoing specification, as indicated the scope of the invention.

We claim:

1. A method for detecting the position of a separator relative to electrodes in a secondary lithium battery comprising the steps of:

providing a secondary lithium battery including a positive electrode, a negative electrode, a X-ray sensitive separator located between the electrodes, and a can housing the electrodes and separator, the X-ray sensitive separator comprising a microporous membrane having a X-ray detectable element dispersed therein, the X-ray detectable element comprising at least 2 and no greater than 20 weight % of the membrane;

subjecting the secondary lithium battery to X-ray radiation;

determining the position of the separator relative to the electrodes; and

approving or rejecting the secondary lithium battery based upon the position of the separator relative to the electrodes.

2. The method of claim **1** wherein the X-ray detectable element comprising 2 to 10 weight % of the membrane.

3. The method of claim **1** wherein the X-ray detectable element comprising 2 to 5 weight o of the membrane.

4. The method of claim **1** wherein the X-ray detectable element comprising about 4 weight % of the membrane.

5. The method of claim **1** wherein the X-ray detectable element being selected from the group consisting of metal, metal oxide, metal phosphate, metal carbonate, X-ray fluorescent material, metal salt, metal sulfate, or mixtures thereof, and any of the foregoing metals being selected from the group consisting of Zn, Ti, Mn, Ba, Ni, W, Hg, Si, Cs, Sr, Ca, Rb, Ta, Zr, Al, Pb, Sn, Sb, Cu, Fe, and mixtures thereof.

6. The method of claim **5** wherein the X-ray detectable element being barium sulfate.

7. A X-ray sensitive battery separator for a secondary lithium battery comprising:

a microporous membrane having a X-ray detectable element at least one of dispersed therein, coated thereon, or added thereto, and said X-ray detectable element comprising at least 2 and no more than 20 weight % of the membrane or separator.

8. The X-ray sensitive battery separator of claim **7** wherein said X-ray detectable element comprising between 2-10 weight % of the membrane.

9. The X-ray sensitive battery separator of claim **7** wherein said X-ray detectable element comprising between 2-5 weight % of the membrane.

10. The X-ray sensitive battery separator of claim **7** wherein said X-ray detectable element comprising about 4 weight of the membrane.

11. The X-ray sensitive battery separator of claim **7** wherein the X-ray detectable element being selected from the group consisting of metal, metal oxide, metal phosphate, metal carbonate, X-ray fluorescent material, metal salt, metal sulfate, or mixtures thereof, and any of the foregoing metals being selected from the group consisting of Zn, Ti, Mn, Ba, Ni W, Hg, Si, Cs, Sr, Ca, Rb, Ta, Zr, Al, Pb, Sn, Sb, Cu, Fe, and mixtures thereof.

12. The X-ray sensitive battery separator of claim **7** wherein the X-ray detectable element being barium sulfate.

13. A secondary lithium battery for X-ray inspection comprising:

a positive electrode, a negative electrode, a X-ray sensitive separator located between the electrodes, and a can housing the electrodes and separator, the X-ray sensitive separator comprising a microporous membrane having a X-ray detectable element dispersed therein, the X-ray detectable element comprising at least 2 and no greater than 20 weight % of the membrane.

14. The secondary lithium battery of claim **13** wherein said X-ray detectable element comprising between 2-10 weight % of the membrane.

15. The secondary lithium battery of claim **13** wherein said X-ray detectable element comprising between 2-5 weight of the membrane.

16. The secondary lithium battery of claim **13** wherein said X-ray detectable element comprising about 4 weight % of the membrane.

17. The secondary lithium battery of claim **13** wherein the X-ray detectable element being selected from the group consisting of metal, metal oxide, metal phosphate, metal carbonate, X-ray fluorescent material, metal salt, metal sulfate, or mixtures thereof, and any of the foregoing metals being selected from the group consisting of Zn, Ti, Mn, Ba, Ni, W, Hg, Si, Cs, Sr, Ca, Rb, Ta, Zr, Al, Pb, Sn, Sb, Cu, Fe, and mixtures thereof.

18. The secondary lithium battery of claim **17** wherein the X-ray detectable element being barium sulfate.

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